

LATTE: A Structure-Informed Latent Model for Protein Sequence Embedding

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Abstract

Background: Deep protein models benefit from embeddings that capture sequence–structure relationships. We introduce LATTE, a structure-aware variational autoencoder trained with a novel structural loss to explicitly learn inter-residue similarity. **Results:** LATTE reconstructs unseen UniRef50 sequences with 97.17% accuracy, remains robust to simulated noise, and generates novel proteins sharing only 10% identity with training data. Its latent space is informative for downstream tasks: Gaussian process models trained on the latent space achieve fivefold CV accuracy of 0.987 for fluorescent-protein (FP) vs non-FP classification and predict excitation/emission peaks with RMSEs of 2.7 nm and 3.8 nm. Compared with ESM-2, pairwise cosine-distance distributions on matched subsets are broader and heavier-tailed for LATTE (mean = 0.1694, SD = 0.3428; p50 = 0.0141, p90 = 0.9006; n = 49,847) versus tighter for ESM-2 (mean = 0.0381, SD = 0.0822; p50 = 0.00953, p90 = 0.1133; n = 49,847). Direct pairwise comparisons yield a regression slope $\mathbf{b} = \mathbf{0.125}$ and intercept $\mathbf{a} = \mathbf{0.017}$ with high rank correlation (Spearman $\rho=0.761$), indicating that LATTE largely preserves neighbor ordering while expanding dynamic range. In $\mathbf{k} = \mathbf{3}$ clustering, cosine-based silhouette scores favor LATTE, and cross-partition agreement indicates strong consistency between the two representations. **Conclusion:** LATTE provides a structure-aware embedding with preserved neighbor ordering and expanded dynamic range, improving latent-space prefiltering prior to alignment (“Deep BLAST”), while supporting accurate reconstruction and effective downstream prediction.

Keywords: protein language model; variational autoencoder; structural loss; fluorescent proteins; embedding geometry; cosine distance; generative modeling; bioinformatics

1 Background

1.1 Protein

Proteins are linear polymers of amino acids linked by peptide bonds, whose three-dimensional conformations determine their biochemical functions. [1–3] The primary structure, the specific sequence of 20 canonical amino acids dictates folding pathways through intramolecular interactions such as hydrogen bonds, hydrophobic packing, electrostatic attractions, and van der Waals forces. These interactions give rise to secondary motifs (α -helices, β -sheets), tertiary folds, and quaternary assemblies in multimeric complexes.

1.2 Deep Learning

Deep learning accelerated protein modeling by learning transferable sequence representations at scale. We defer architectural details to Methods and focus here on the motivation and role of the latent space in design.

1.3 ESM-2 and ESMS

Using pretrained ESM-2 embeddings, we developed ESMS that reduces inference time relative to ESM-2’s 0.5 s per sequence while preserving high embedding fidelity. ESM-2, introduced by Rives et al., is a masked language model that captures short- and long-range residue interactions [4], and it underpins ESMFold, a three-dimensional structure predictor with performance comparable to AlphaFold 2 [5, 6], and is widely used for downstream tasks such as secondary-structure and thermostability prediction. Although available at multiple parameter scales (650M parameters being most common), ESM-2’s latency motivates lighter alternatives. On a held-out test set, ESMS achieved cosine similarity of 0.9647 and RMSE of 1.2998.

We position our approach within recent progress on large protein language models. In particular, ESM-3 tightly couples sequence, structure, and function supervision and demonstrates design/editing capabilities beyond prior ESM-2 systems [7]. LATTE is intended as a lightweight, structure-informed generator that complements such foundation models by emphasizing controllable latent structure and efficient training.

1.4 BLAST

BLAST (Basic Local Alignment Search Tool) locates local similarities via a *seed-and-extend* heuristic, quickly forming high-scoring segment pairs (HSPs) [8]. The statistical significance of alignments is modeled by the Karlin–Altschul framework, yielding the *E*-value and bit score [9]. While less sensitive than optimal dynamic-programming alignment, BLAST scales effectively to large databases and remains the practical standard; the BLAST+ re-architecture further streamlined performance and modularity [10].

1.5 Segment Tree (Generic Prefilter)

A segment tree hierarchically partitions a 1D domain to answer range queries (e.g., sum, min/max) and support point updates in $O(\log n)$ time [11, 12]. As a generic *prefilter*, it can prune large candidate sets using bounds or aggregated statistics before any high-cost procedure (e.g., precise alignment) is applied. In this work we rely only on standard properties and complexities of the segment tree; implementation variants are out of scope.

1.6 Our Contribution

It has been shown that transformer architectures are capable of capturing hidden information in a sequence. Also, rather than training big transformer models, using a pretrained model as an embedding was demonstrated to be effective. However, only training with a sequence has clear limits on generalization, function, and versatility. LATTE, to overcome such limits, introduced a custom loss function that explicitly forces it to learn and represent structural information in the latent vector space.

2 Methods

2.1 Shaping the latent space

We trained LATTE with a pseudo-decoder which acts as a training-signal generator that receives encoder memory and latent vectors. Moreover, we enforce structural consistency via a perceptual loss computed from pretrained ESMS embeddings, which capture residue–position regularities at substantially lower cost than full structure predictors. Given an original sequence $\mathbf{x}_{\text{orig}}$ and its reconstruction $\mathbf{x}_{\text{recon}}$, we form two embedding-based terms.

$$L_{\text{COS}} = \left[1 - \cos(\text{ESMS}(\mathbf{x}_{\text{orig}}), \text{ESMS}(\mathbf{x}_{\text{recon}})) \right], \quad (1)$$

$$L_{\text{MSE}} = \|\text{ESMS}(\mathbf{x}_{\text{orig}}) - \text{ESMS}(\mathbf{x}_{\text{recon}})\|_2^2. \quad (2)$$

These structural terms are combined with next-token cross-entropy L_{CE} (teacher forcing) and KL divergence L_{KL} into

$$L_1 = \lambda(L_{\text{COS}} + L_{\text{MSE}}) + \alpha L_{\text{CE}} + \beta L_{\text{KL}} \quad (3)$$

with $\lambda = 5$, α linearly decayed from 30 to 0.1 over the first 100 epochs, and β linearly annealed from 0 to 0.1 over the same interval. The cosine term tolerates substitutions among similarly likely residues, while the MSE term penalizes large deviations. Together, they discourage posterior collapse by making accurate reconstruction contingent on informative latents. [13].

LATTE is a comparably lightweight transformer with 5.5M parameters, composed of 4-layer transformer encoders (Figure 1). The hyperparameters are detailed in Table 1. The Adam optimizer [14] was used.

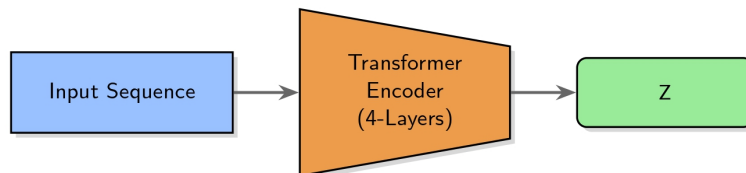


Fig. 1: The architecture of LATTE. An input sequence is processed by a 4-layer Transformer Encoder, which outputs the parameters (mean and log-variance) of the latent distribution. A latent vector is sampled and then reconstructed into the output sequence by a 4-layer Transformer Decoder.

Table 1: Hyperparameters for LATTE.

	Vocab Size	d_model	Latent Dim	n_heads	Feed Forward	Dropout
Value	33	256	256	4	512	0.3

2.2 Implementation and Training

Model selection favored the epoch 380 checkpoint (LATTE _ 380), which balanced the KL divergence near the active threshold (0.048) and achieved the lowest validation cross-entropy loss. The model was trained using two T4 GPU sessions provided by Kaggle on a random subsample of the UniRef50 dataset. Monitored learning curves did not show any signs of overfitting (Supplementary Figure S1). Although the epoch 500 model (LATTE _ 500) reached a reconstruction rate of 99.976% with validation losses of Val CE = 0.000, COS = 0.003, MSE = 0.007, and KL = 0.002, its very low KL value suggested potential posterior collapse (Supplementary Figures S2a and S2b). Adding noise to the latent space did not significantly affect reconstruction performance. Instead, epoch 380 was chosen because its KL divergence (0.048) was closer to the recommended active value of 0.05, and it exhibited the lowest Val CE. At epoch 380, validation losses were Val CE = 0.072, COS = 0.010, MSE = 0.020, and KL = 0.048 (Supplementary Figures S3a and S3b). Figure 2(a) and (b) illustrate the epoch-wise progression of CE and KL values, confirming that epoch 380 offers the optimal balance.

3 Results

LATTE’s latent space remains active and enables true de novo sequence generation, achieving a mean KL divergence per dimension of 0.04998 (RMSE 0.07027). We confirmed absence of posterior collapse by sampling random latent vectors values around 0.05 to ensure decoder utilization and comparing decoded sequences against originals, demonstrating that LATTE produces genuinely novel protein sequences rather than memorizing the training data.

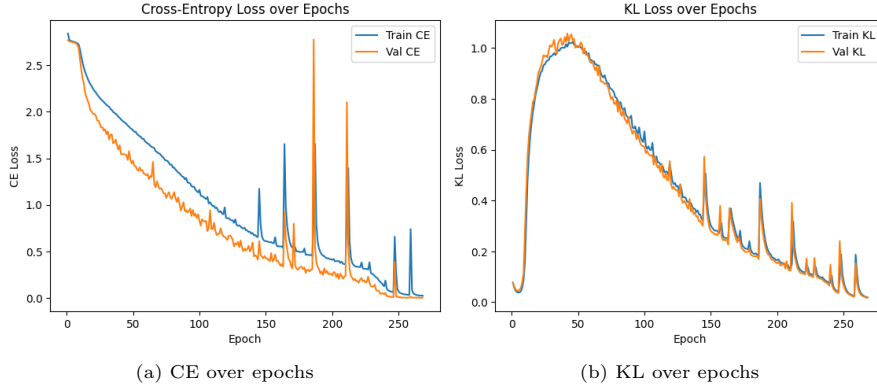


Fig. 2: CE and KL values over epochs with structural loss.

Table 2: GFP spectral-prediction pipelines on different embeddings.

Embedding	Classifier metric (5-fold)	λ_{abs} RMSE (nm)	λ_{em} RMSE (nm)
ESM-2 (650M)	AUC 0.997	2.70	3.80
LATTE latent (256-d, $\sim 5.5\text{M}$)	AUC 0.987	2.70	3.80

3.1 Case Study: Fluorescent Protein (FP) Analysis and Generation

We used pretrained ESM-2 embeddings. Using manually curated FP sequences from FPbase [15], we trained Gaussian process [16] (GP) models to (1) classify FPs versus non-FPs and (2) regress their spectral peaks. The GP classifier achieved 0.987 accuracy under fivefold cross-validation, while the GP regressor achieved root-mean-square errors of 2.70 nm for maximum absorption (A_{abs}) and 3.80 nm for emission (λ_{em}). Classification reports on both training and test sets confirm strong performance without signs of overfitting (Tables 3 and 4).

ESM-2 baseline and comparison. With the same Gaussian process (GP) pipelines, an *ESM-2 (650M)* embedding baseline produced strong discriminative and regression performance: 5-fold AUC 0.997 and wavelength RMSEs of 2.70/3.80 nm for $\lambda_{\text{abs}}/\lambda_{\text{em}}$. Using *LATTE*’s 256-d latent embeddings, the FP vs. non-FP classifier reached 5-fold accuracy 0.987. This indicates that *LATTE*’s compact latent space provides competitive downstream signal at roughly two orders of magnitude fewer parameters and substantially lower inference cost, while ESM-2 serves as an upper-bound reference for fully supervised spectral-peak regression.

The latent space visualization via t-SNE clearly separates fluorescent from non-fluorescent proteins and encodes spectral properties as continuous gradients, confirming that structural information is captured effectively [17] (Supplementary Figure

Table 3: Classification report on the training set.

	Precision	Recall	F1-score	Support
Non-FP (0)	0.9920	0.9940	0.9930	501
FP (1)	0.9880	0.9840	0.9860	250
Accuracy			0.9907	751
Macro Avg	0.9900	0.9890	0.9895	751
Weighted Avg	0.9907	0.9907	0.9907	751

Table 4: Classification report on the test set.

	Precision	Recall	F1-score	Support
Non-FP (0)	0.9840	0.9840	0.9840	125
FP (1)	0.9683	0.9683	0.9683	63
Accuracy			0.9787	188
Macro Avg	0.9761	0.9761	0.9761	188
Weighted Avg	0.9787	0.9787	0.9787	188

S4). Non-fluorescent proteins cluster on the inner side of two curves, whereas fluorescent proteins occupy the outer region. Moreover, color gradients corresponding to emission and absorption wavelengths indicate that proteins with similar spectral peaks are grouped together.

k-means clustering of the projected FP embeddings revealed three distinct clusters (Table 5), whose consensus vectors were decoded into novel sequences truncated to each cluster’s mean length. Classification of these generated sequences using the trained GP classifier showed that Cluster 1 achieved a 100% success rate (Supplementary Figure S5), suggesting that intra-cluster sample quality and consistency are more critical than sample size for producing functional proteins. The generated proteins exhibited 75% - 96% identity with FPbase sequences, reflecting the high similarity of their latent representations.

Table 5: Statistics for FP clusters identified by k-means.

Cluster	n (samples)	Mean Length	Std (Length)
0	22	316	5.28
1	318	234	10.00
2	14	118	29.71

Close vectors in the latent space encode proteins with similar three-dimensional structures, as confirmed by AlphaFold predictions. Cluster 1 proteins exhibited classic

Table 6: Pairwise cosine distance ($1 - \text{cosine similarity}$) summary for LATTE vs. ESM-2 embeddings.

Embedding	n	Mean	SD	p_{10} / p_{50} / p_{90}
LATTE (latent)	49,847	0.1694	0.3428	0.00257 / 0.01406 / 0.90065
ESM-2 (650M)	49,847	0.0381	0.0822	0.00365 / 0.00953 / 0.11329

β -barrel folds with pLDDT scores above 90; Cluster 0 proteins formed two groups of three internal β -strands wrapped by α -helices; and although Cluster 2 proteins did not adopt β -barrels, they nonetheless shared a common fold of three β -strands surrounded by α -helices.

Three candidate sequences were selected for in vitro validation two from Cluster 1 and one from Cluster 2 based on GP classifier predictions, structural similarity to the original proteins, and suitability for alternative fluorescent molecules. The full amino-acid sequences of these candidates are provided in Supplementary Data S1.

3.2 Embedding geometry of LATTE vs. ESM-2 (cosine distance)

To compare the global structure of the embedding spaces, we computed pairwise cosine distances ($1 - \text{cosine similarity}$) on matched subsets and summarized both the distributions and their empirical cumulative distribution functions (ECDFs). LATTE latent distances are broader with a heavier tail (mean = 0.1694, SD = 0.3428, $p_{50} = 0.0141$, $p_{90} = 0.9006$; $n = 49,847$), whereas ESM-2 distances are tighter (mean = 0.0381, SD = 0.0822, $p_{50} = 0.00953$, $p_{90} = 0.1133$; $n = 49,847$) (Figure 3, Table 6). This suggests that ESM-2 clusters points more compactly, while LATTE yields a more diffuse geometry that can increase recall for remote neighbors; in practice, we combine LATTE’s structure-aware prefilter with sequence alignment (“Deep BLAST”) to recover interpretability and precision.

Pairwise cosine-distance matrices for the 354-sequence subset in the two embedding spaces. Left: ESM; right: LATTE. Brighter horizontal/vertical bands mark sequences or clusters with larger distances to many others, highlighting global structure and potential outliers. (Figure 5)

To directly compare the pairwise geometries, we plotted ESM (y) versus LATTE (x) cosine distances for matched pairs (Figure 4). Distances are systematically smaller in ESM (slope $b=0.125$; intercept $a=0.017$), yet the rank correlation remains high (Spearman $\rho=0.761$), supporting that LATTE preserves ESM’s neighbor ordering while expanding the dynamic range, a property we exploit for latent prefiltering in Deep BLAST.

We report a rerun of $k=3$ clustering to compare LATTE and ESM-2 embeddings. Silhouette scores (cosine) favor LATTE, and cross-partition agreement metrics indicate strong consistency between the two representations[18].

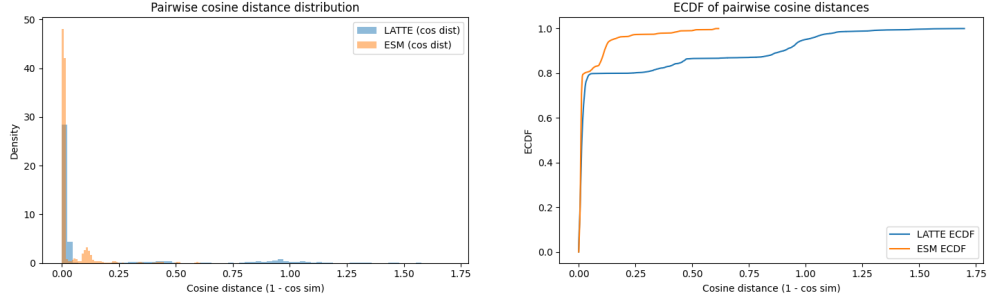


Fig. 3: Pairwise cosine distance distributions (left, density histograms) and ECDFs (right) for LATTE latent vs. ESM-2 embeddings.

Table 7: Silhouette (cosine), $k=3$.

Metric	LATTE	ESM-2
Silhouette (cosine)	0.9431	0.9022

Table 8: Summary of $k=3$ clustering agreement.

Metric	LATTE	ESM-2
Cross-partition agreement (LATTE vs. ESM-2)		
Adjusted Rand Index (ARI)		0.7373
Adjusted Mutual Information (AMI)		0.6055
Fowlkes–Mallows Index (FMI)		0.9596
Variation of Information (VI)		0.3529
Purity (LATTE $\rightarrow$ ESM-2)		0.9633

3.3 Ablation Study

An ablation study was conducted with the loss function

$$L = \alpha CE + \beta KL,$$

where the weights are set to $\alpha = 30, \beta = 0$ for epochs ≤ 30 , and $\alpha = 0.1, \beta = 0.1$ thereafter. In this setup, KL vanishing occurred right after epoch 100, demonstrating the crucial role of the structural loss term.

Figure 6 shows the CE and KL curves when no structural loss is used, and Figure 2 shows the corresponding curves with structural loss incorporated. Only the latter prevents KL collapse and yields more stable convergence.

4 Discussion and future directions

Enforcing structural integrity via AlphaFold-derived losses or contact maps is impractical at scale: AlphaFold requires 10 to 20 minutes per sequence, contact maps demand

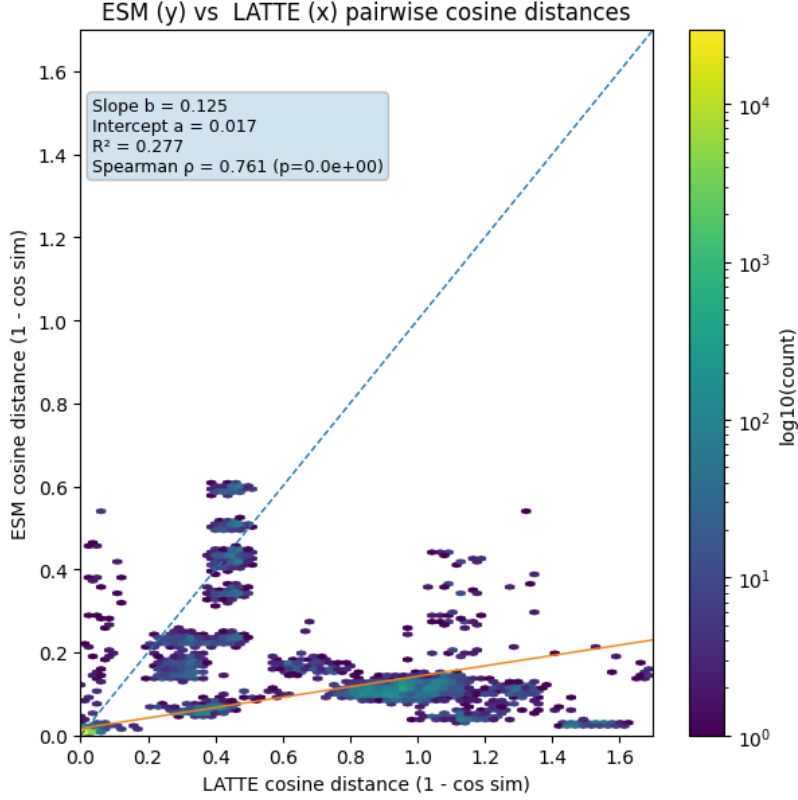


Fig. 4: ESM (y) vs. LATTE (x) pairwise cosine distances for matched pairs. The dashed line marks $y=x$; the solid line is an OLS fit ($b=0.125$, $a=0.017$; $R^2=0.277$). Spearman $\rho=0.761$ indicates strong rank concordance despite scale compression in ESM.

Table 9: Confusion matrix (rows: LATTE-aligned clusters; columns: ESM-2 clusters).

	ESM-2-0	ESM-2-1	ESM-2-2
LATTE-0	0	0	1
LATTE-1	0	23	1
LATTE-2	6	5	318

large memory and rely on limited known structures and require aligning for calculation. A more scalable approach is to leverage other protein language models (PLMs) such as ProTrans5 or ProtBert, ideally their smaller variants like ESM or ESM-2, for structural supervision.

We propose Deep BLAST: retrieve a small candidate set via LATTE’s 256-d, structure-informed latent embeddings (cosine similarity), then run classical BLAST

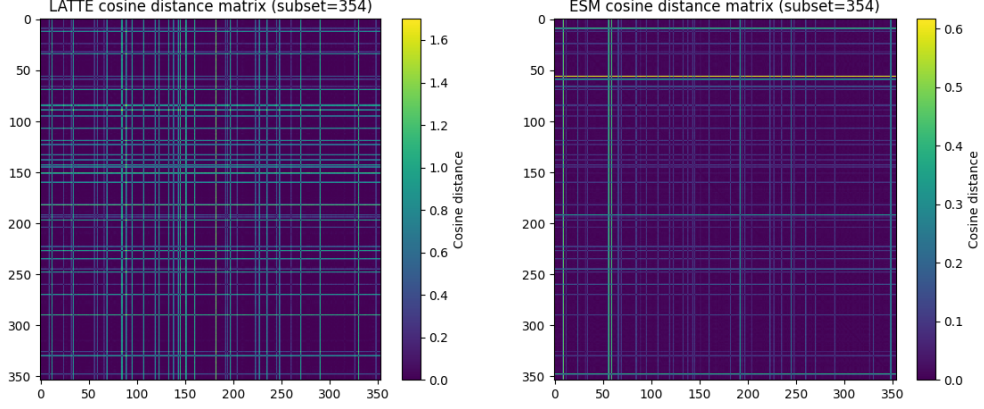


Fig. 5: Pairwise cosine-distance matrices for the 354-sequence subset in the two embedding spaces. Left: LATTE; right: ESM.

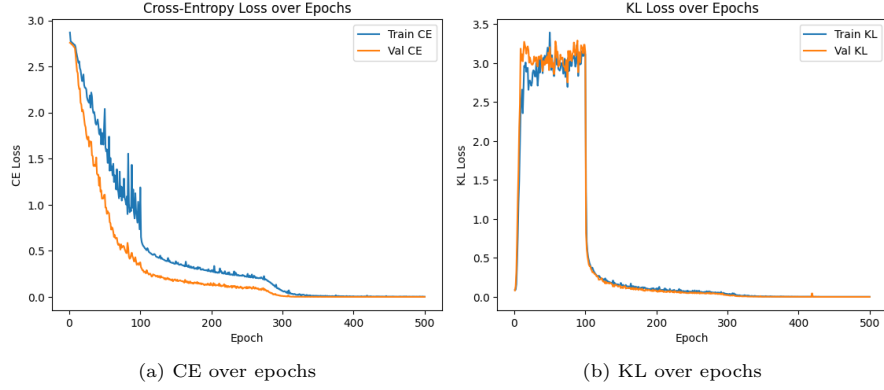


Fig. 6: CE and KL values over epochs for the ablation study model without structural loss.

only on that shortlist. Conventional BLAST depends on fixed seed/word heuristics, so “nearest alignable” hits need not be nearest in function/structure, and remote homology can be missed or diluted by low-complexity/repeats. Latent-space prefiltering enriches for biologically coherent neighbors and reduces the search fan-out by orders of magnitude (e.g., 10–100 $\times$, tunable K/M), shifting BLAST from broad discovery to precise refinement. In practice, this yields higher precision at top ranks and lower wall-clock/CPU cost while preserving interpretability via the downstream alignment. A conservative fallback triggers standard BLAST when latent similarity falls below a preset threshold.

Table 10: Per-cluster sizes and reported means (μ).

Cluster	size	LATTE μ	ESM-2 μ
0	6	0.000	0.728
1	28	0.904	0.737
2	320	0.949	0.920

5 Conclusion

LATTE’s GFP case study shows that a compact, structure-aware latent space’s 256 dimensions with an active KL near 0.05 captures the functional and spectral determinants of fluorescent proteins without relying on sequence generation. A Gaussian process classifier trained on LATTE embeddings separates FPs from non-FPs with 0.987 fivefold cross-validation accuracy, and a GP regressor predicts excitation and emission peaks with RMSEs of 2.70 nm and 3.80 nm, respectively. Manifold visualizations (t-SNE) reveal clear FP/non-FP separation and smooth spectral gradients across the latent manifold, consistent with embeddings that encode structural constraints.

Taken together, these results indicate that LATTE’s low-dimensional latent representation is informative and structure-aligned, enabling reliable property inference and providing a general path to extend property-guided analysis to other functional families.

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Declarations

Ethics approval and consent to participate

This study involved only computational analyses on publicly available protein sequence datasets and did not involve human participants, animals, or identifiable personal data. Ethics approval and consent to participate were therefore not required.

Consent for publication

Not applicable.

Availability of data and materials

The fluorescent protein dataset is available at FPbase: <https://www.fpbase.org/> [19]. A copy is also hosted on Kaggle: [dannyahn/fluorescent-protein-dataset](https://www.kaggle.com/dannyahn/fluorescent-protein-dataset) [20]. The source code and trained models are publicly available on GitHub: <https://github.com/Ahnd6474/LATTE> [21].

Competing interests

The authors declare that they have no competing interests.

Funding

This work was supported by the Daegu Science High School student independent research program. No specific grant number was assigned.

Authors' contributions

CRedit taxonomy: DA: Conceptualization, Methodology, Software, Writing—original draft, Project administration

Acknowledgements

We thank Minjae Lee, Sihyeon Moon, and Jooyoung Jung for their assistance with manuscript proofreading and for organizing the GitHub data repository.